Session 8: Machine Learning Foundations & Practical PyTorch

KB8029 — Bioinformatics & Deep Learning

Arne Elofsson

SciLifeLab

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Log in to the lecture quiz

Log in to the quiz

Scan the code, or go to quiz.bioinfo.se/live

Log in with your university account

Answering counts as being at the lecture: up to 3 bonus points on the exam

Right or wrong is never graded, only whether you answered. Results are shown without names.

quiz.bioinfo.se/live

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Warm-up: what do you know already?

3 quick questions

On the topics of today, one at a time, about 30 seconds each

Answer on your own, and guess if you do not know

It shows me where to start; right or wrong is never graded

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Predicting a protein's melting temperature from its sequence is ...

A classification

B regression

C clustering

D unsupervised learning

Answer now

Scan the code, or go to

quiz.bioinfo.se/live

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95 of 100 proteins are not membrane proteins. A model always says “not membrane”. What is its accuracy?

A 0%

B 5%

C 50%

D 95%

Answer now
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quiz.bioinfo.se/live

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In a PyTorch training step, what does loss.backward() do?

A Updates the weights

B Computes the gradients

C Computes the loss

D Sets the gradients to zero

Answer now
Scan the code, or go to
quiz.bioinfo.se/live

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What the room knows already

Where we start

Which topics most of you know, and which we take slowly today.

Show the results ▸ quiz.bioinfo.se/live/report

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Your questions before today

What you asked

The questions you sent in on Canvas before the lecture

The most common ones now; the others in the block where they belong

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In up to three words: what do you want to learn today?

Posing the problem	Did it learn?	Towards networks
regression or classification	splits, metrics, homology leakage	softmax, nn.Module, parameters

Answer now

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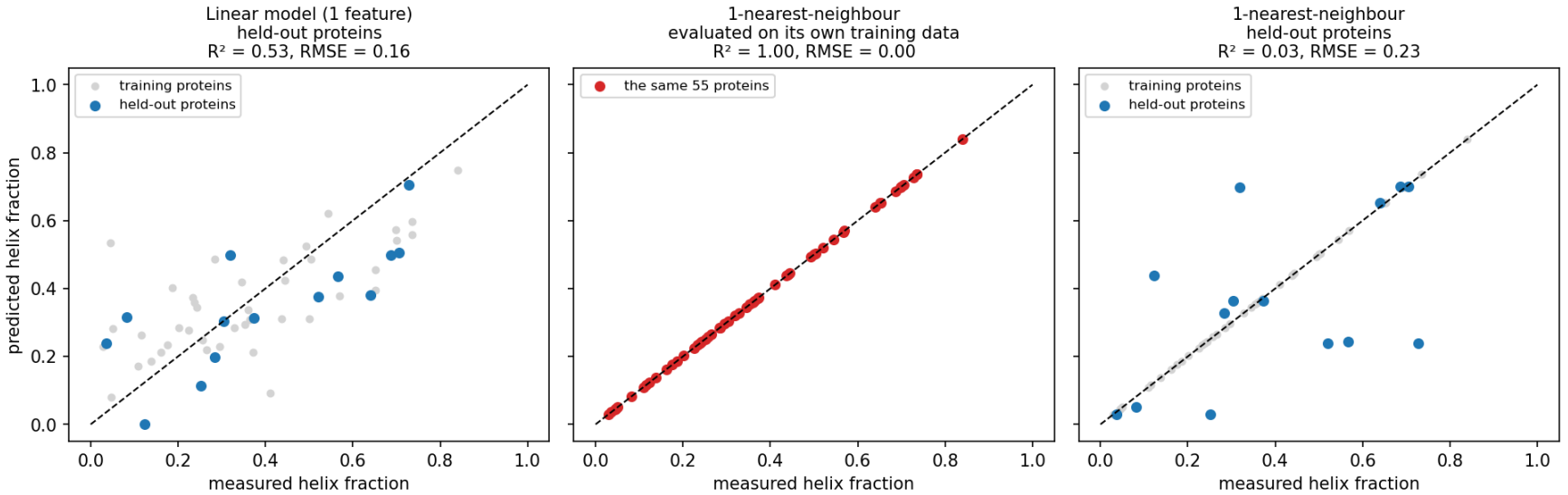
What you want to learn today

What the room wants to learn

Show the results ▸ quiz.bioinfo.se/live/report

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How do we know a model learned anything useful?



The same kind of model, two answers. Middle: 1-nearest-neighbour evaluated on its own training data looks perfect (R^2 1.00). Right: on 14 proteins it has not seen, it is no better than predicting the mean (R^2 0.03).

Notebook: session08-ml-foundations-and-pytorch.ipynb, 'Evaluating a regression model'.

Three blocks: about 30 min talk and 15 min discussion



Block 1: posing the problem (regression or classification). Block 2: evaluation and generalisation, the heart of today. Block 3: towards training networks (activations, softmax, PyTorch).

Times are approximate: a 10 min start (warm-up quiz and your questions), two 15 min breaks and a 5 min close. Each block ends with a quiz question: guess, run the notebook and work out why it behaves as it does, discuss, answer again.

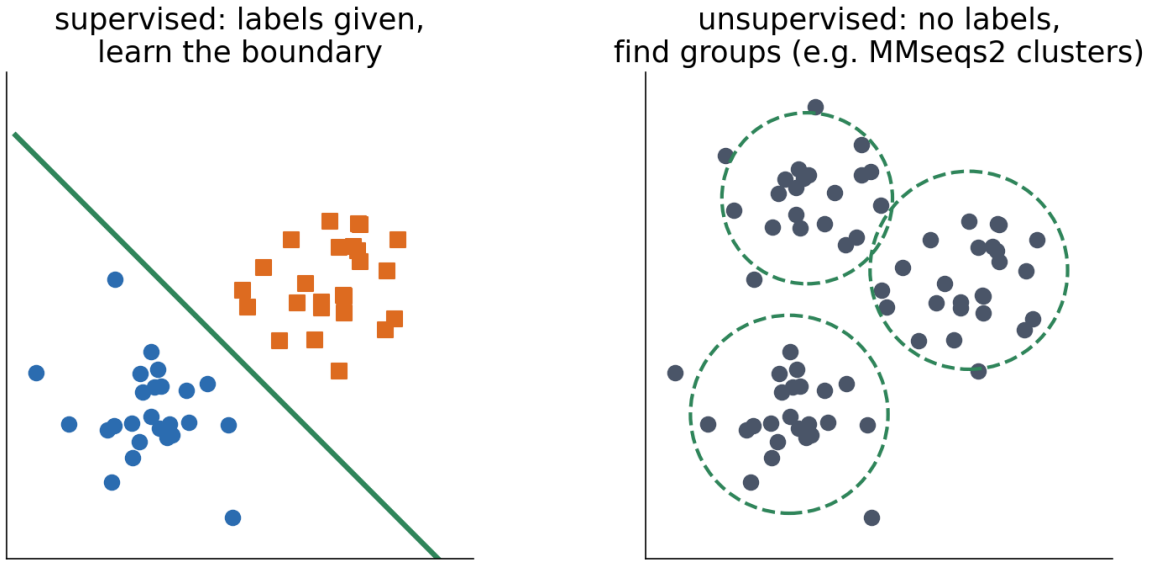
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BLOCK 1 · ABOUT 30 MIN

Posing the problem

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Supervised and unsupervised learning



illustrative data

Supervised: fit a model to labelled examples (a perceptron on known binding sites, Session 7). Unsupervised: find structure without labels (clustering sequences into families). Almost everything we train in this course is supervised.

See course.bioinfo.se/session08.html, 'Supervised and unsupervised learning'.

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The label decides: regression or classification

Regression

the label is a number

- Helix fraction
- Melting temperature, ΔΔG of a mutation
- Expression level, binding affinity (pKd)

Classification

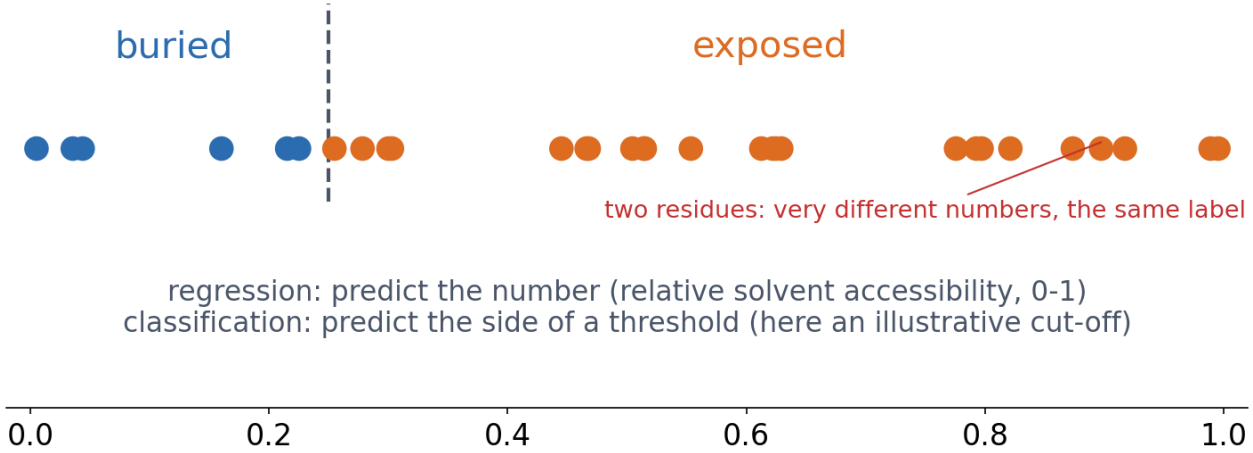
the label is a category

- Subcellular compartment
- Pathogenic or benign? Signal peptide?
- Helix / strand / coil per residue

See course.bioinfo.se/session08.html, Table 8.1.

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Same biology, posed either way

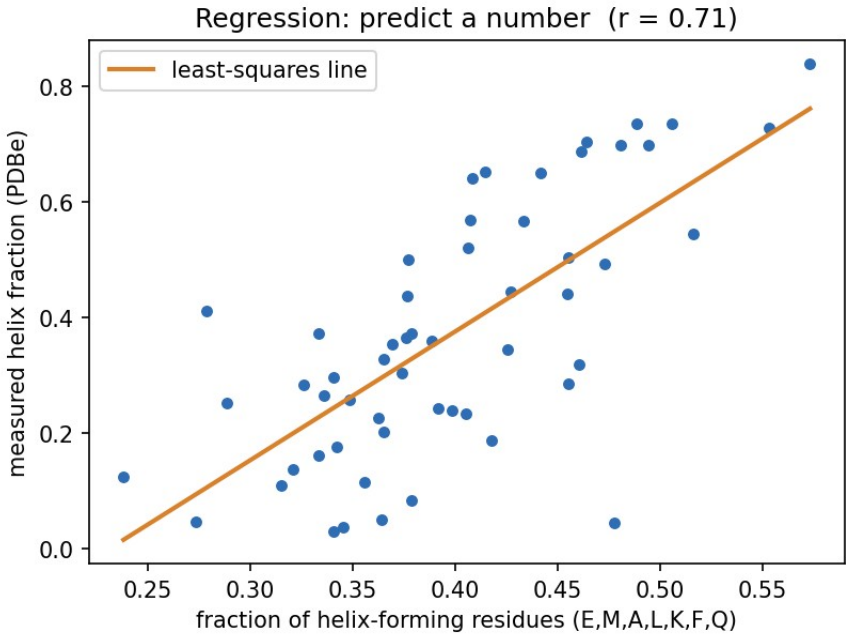


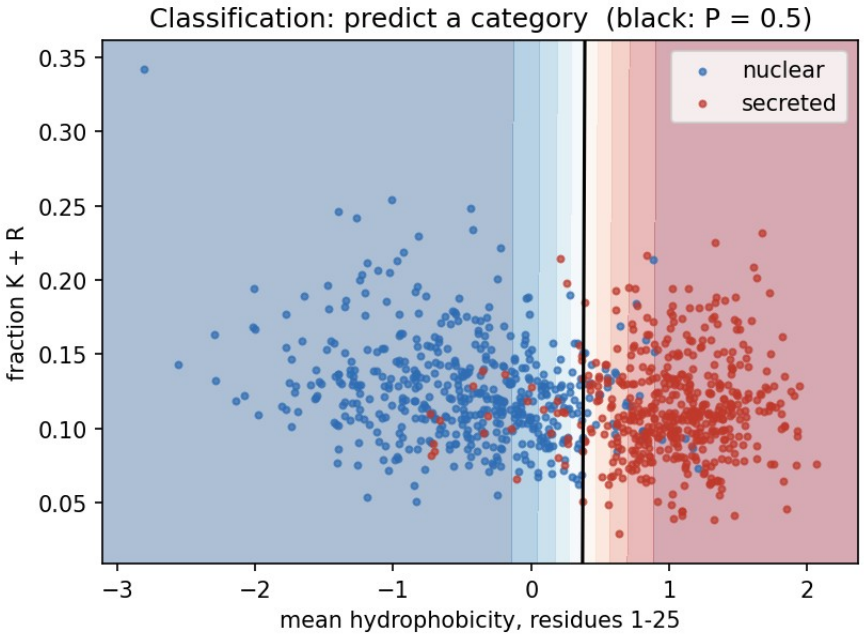
- **A threshold turns regression into classification**
- Easier problem
- **But information is thrown away**

Illustrative values. See course.bioinfo.se/session08.html, 'Regression vs. classification'.

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A regression and a classification, from real data





Left: helix fraction of 55 PDB entries against helix-former content (r = 0.71). Right: 488 secreted and 497 nuclear proteins by N-terminal hydrophobicity and K+R content; logistic regression draws the boundary at P(secreted) = 0.5.

Data fetched live from PDBe and UniProt; notebook session08-ml-foundations-and-pytorch.ipynb.

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Signal peptides: overfitting is as old as the field

- **A hydrophobic N-terminal address: 'send me into the secretory pathway'**
- 1983: a cleavage-site rule, 92% on its own data, 64% on new data
- 1985, McGeoch: two features, boundary drawn by hand
- 1991: a network that fitted every training protein, then failed on new ones
- **Ours: the same two-feature idea, boundary learned, tested on unseen proteins**

92%

1983 rule, on its own 78 sequences

64%

the same rule, on new data

Nielsen, Tsirigos, Brunak & von Heijne (2019), Protein J 38:200; McGeoch (1985), Virus Res 3:271; Ladunga et al. (1991), Bioinformatics 7:485.

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The classifier found the signal peptide

- **Secreted proteins start with a hydrophobic signal peptide**
- The model found that one feature almost on its own
- **93.5% correct on held-out proteins, from two numbers**

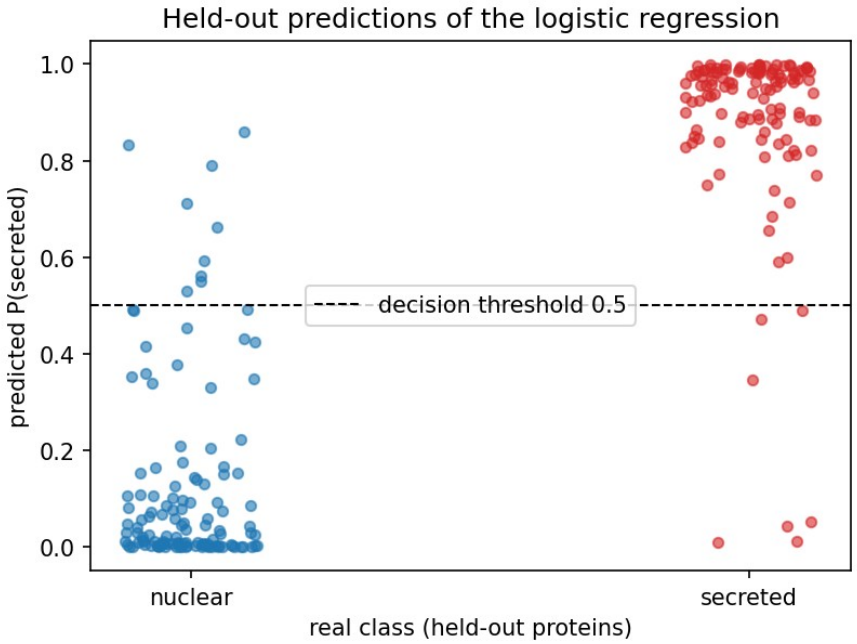
4.26
weight: N-terminal hydrophobicity

−0.20
weight: K+R content

Notebook, 'A classification problem: secreted or nuclear?'.

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Almost every held-out protein falls on the right side

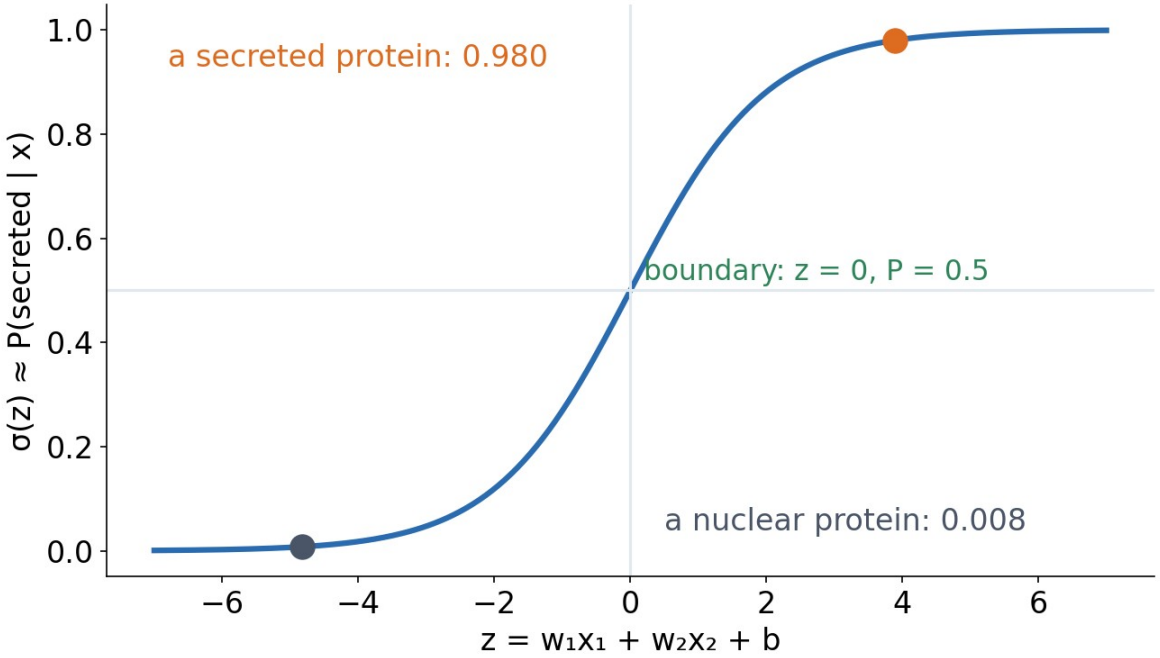


Predicted P(secreted) for each held-out protein, grouped by its real class. The few on the wrong side of 0.5 are the classifier's errors.

Notebook, 'A classification problem: secreted or nuclear?'.

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Logistic regression: a perceptron with a smooth output



- **Same weighted sum as Session 7**
- $\sigma(z) = 1 / (1 + e^{-z})$ maps to (0, 1)
- Boundary: still a straight line
- Smooth, so trainable (Session 9)

See course.bioinfo.se/session08.html, 'Logistic regression: the bridge between the two'.

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What follows from the choice

Regression

linear output

- Loss: mean squared error
- Metrics: RMSE, Pearson r, R^2

Classification

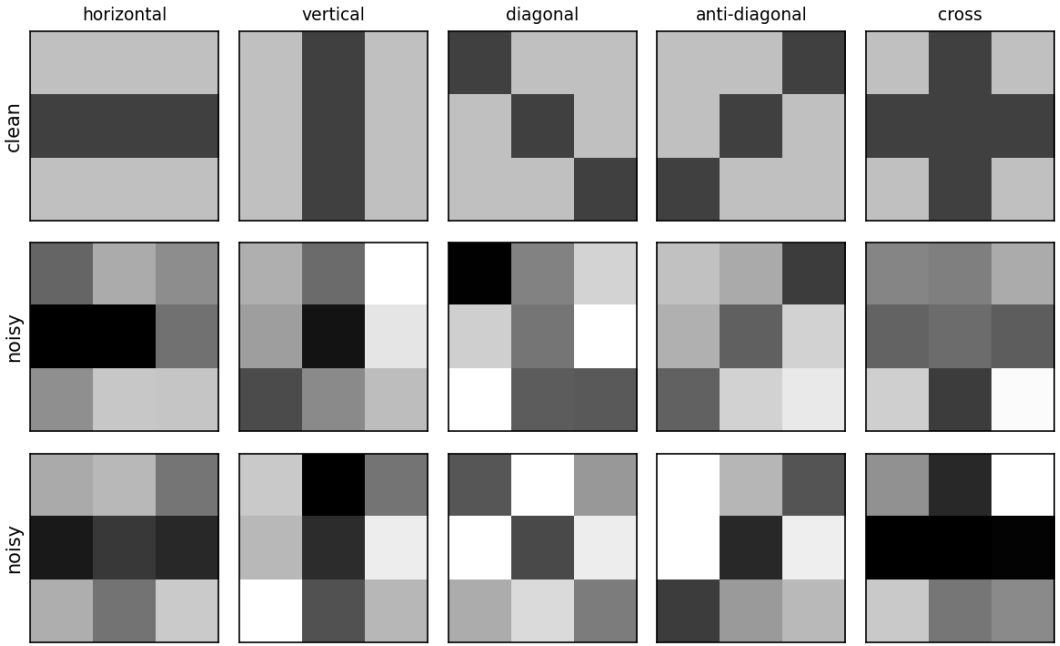
sigmoid / softmax output

- Loss: cross-entropy (Session 1)
- Metrics: confusion matrix, precision, recall, F1, MCC, AUROC

See course.bioinfo.se/session08.html, Table 8.2. Losses are Session 9's topic.

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The smallest image problem: 3×3 pixel patterns

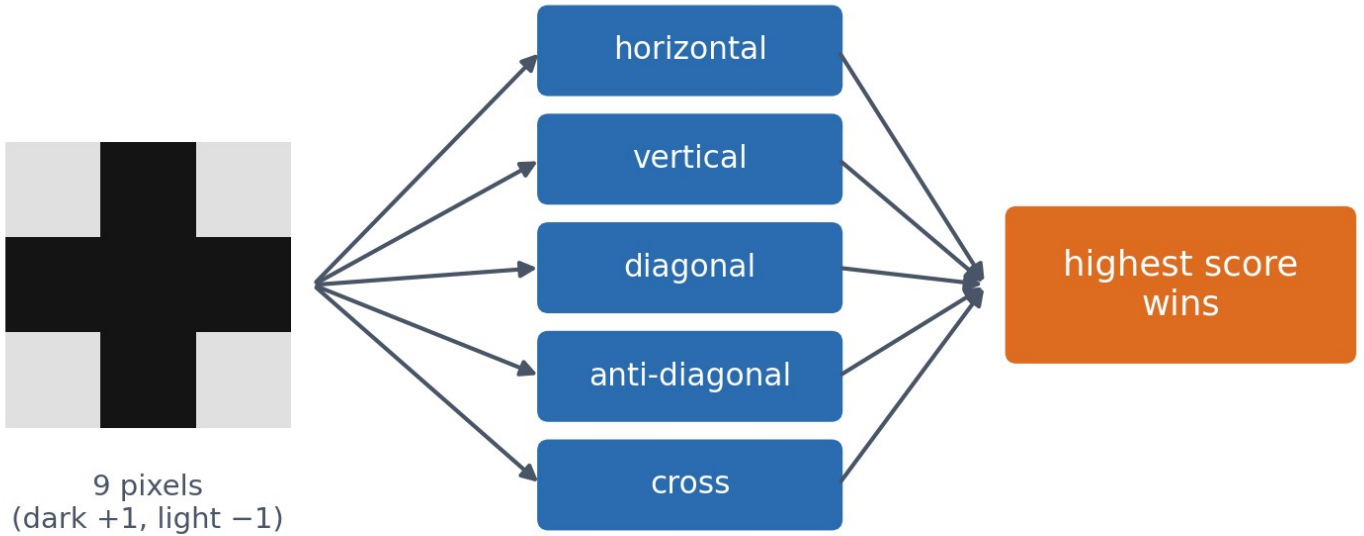


- **9 pixels, 5 classes**
- Clean pattern + noise (σ = 0.8)
- 'Cross' shares pixels with two others

Idea from Ekman, Learning Deep Learning, ch. 2.

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Five perceptrons, one per pattern



five perceptrons, each trained: +1 for its own pattern, −1 for all others (one-vs-rest)

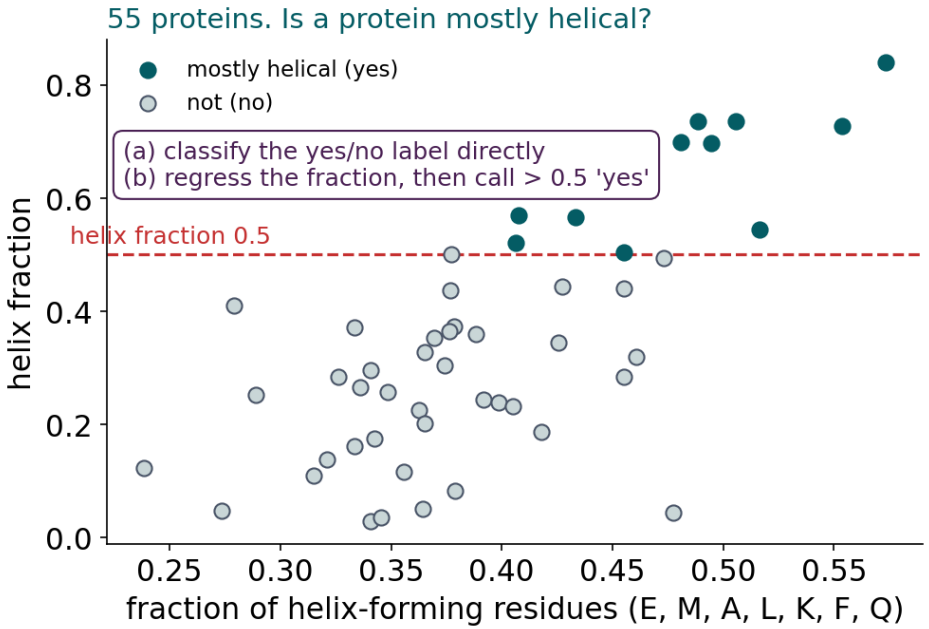
One-vs-rest: each perceptron outputs +1 for its own pattern and −1 for every other; the predicted class is the highest score. No hidden layer, so no backpropagation needed.

See course.bioinfo.se/session08.html, 'A running toy example: 3×3 pixel patterns'.

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Classify the label, or regress the number?

55 proteins; is each mostly helical (helix fraction > 0.5)? One feature: the share of helix-forming residues. (a) Train a classifier on the yes/no labels, or (b) train a regression on the helix fraction and call > 0.5 'yes'. Which is more accurate?



A (a) Classifier: trained on yes/no

B (b) Regression: keeps more information

C About the same

D Neither: one feature is too little

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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Classify the label, or regress the number?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

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Classify the label, or regress the number?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session08-discussion-1.ipynb

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Classify the label, or regress the number?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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Classify the label, or regress the number?

Results, and who explains

Both rounds side by side.

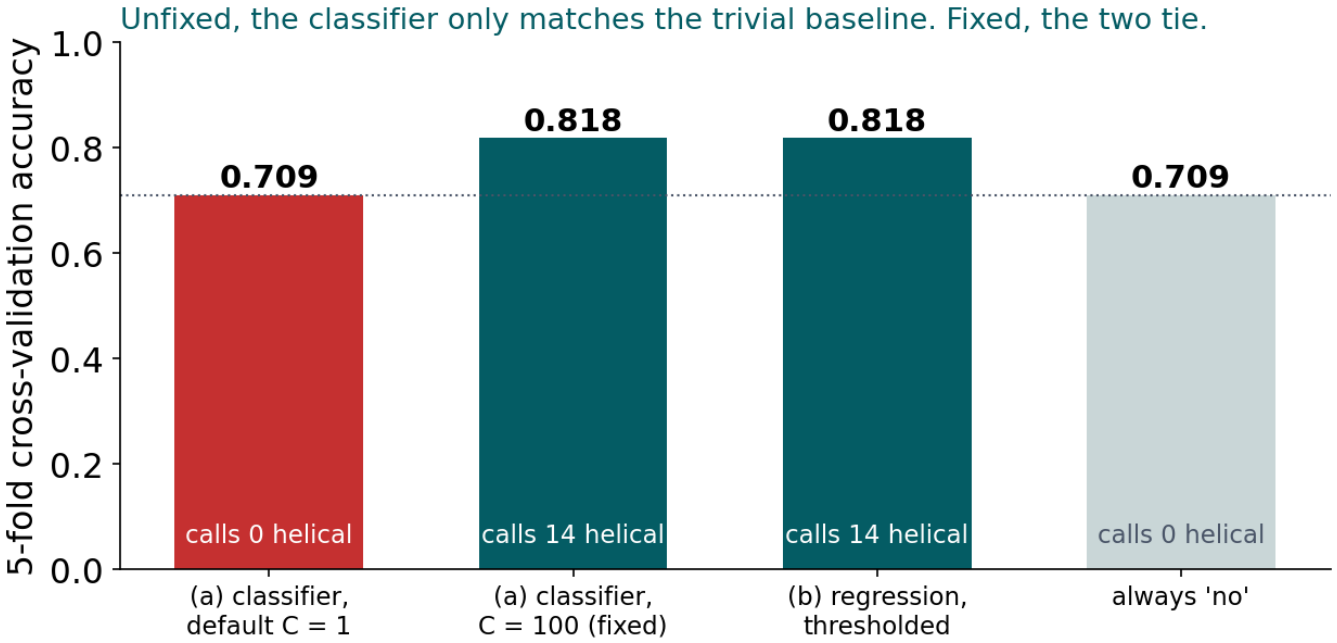
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 1 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session08-discussion-1.ipynb

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The answer: about the same, once the classifier is fixed



With scikit-learn's default regularisation the classifier never says 'helical': its 0.709 is exactly the score of always answering 'no'. With weaker regularisation (C = 100) it reaches 0.818, the same as the regression. Always compare with the trivial baseline; and on 55 proteins neither approach is better.

BLOCK 1 DONE

Break

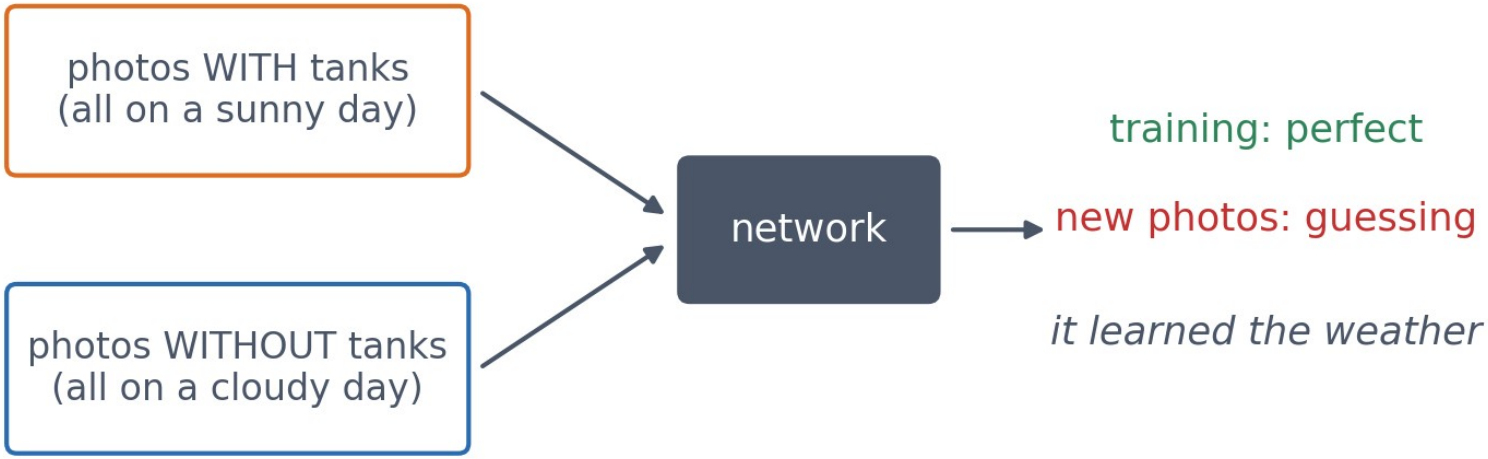
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BLOCK 2 • ABOUT 30 MIN

How do we know a model learned anything?

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Datasets should be representative



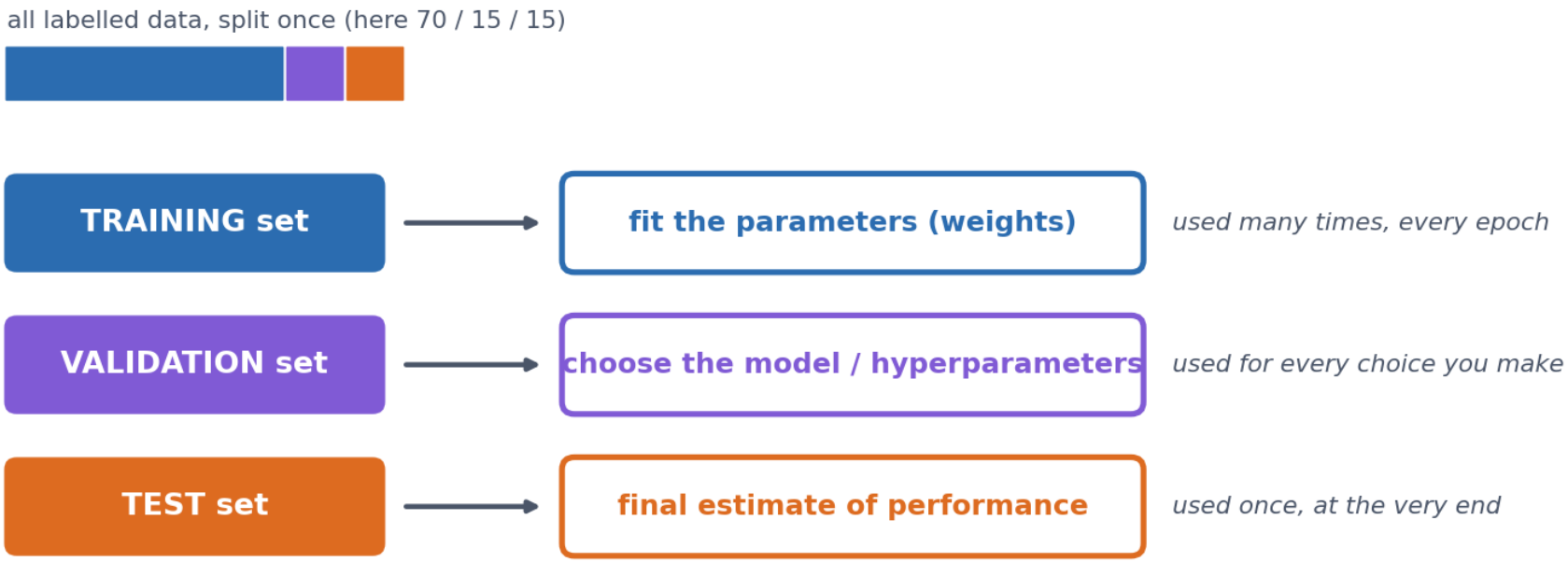
an often-told story, perhaps partly legend; the failure is real and common

The network separated the training photos perfectly and was no better than guessing on new ones: every tank photo was sunny, every empty one cloudy.

See course.bioinfo.se/session08.html, callout 'Datasets should be representative'.

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Why three data sets?

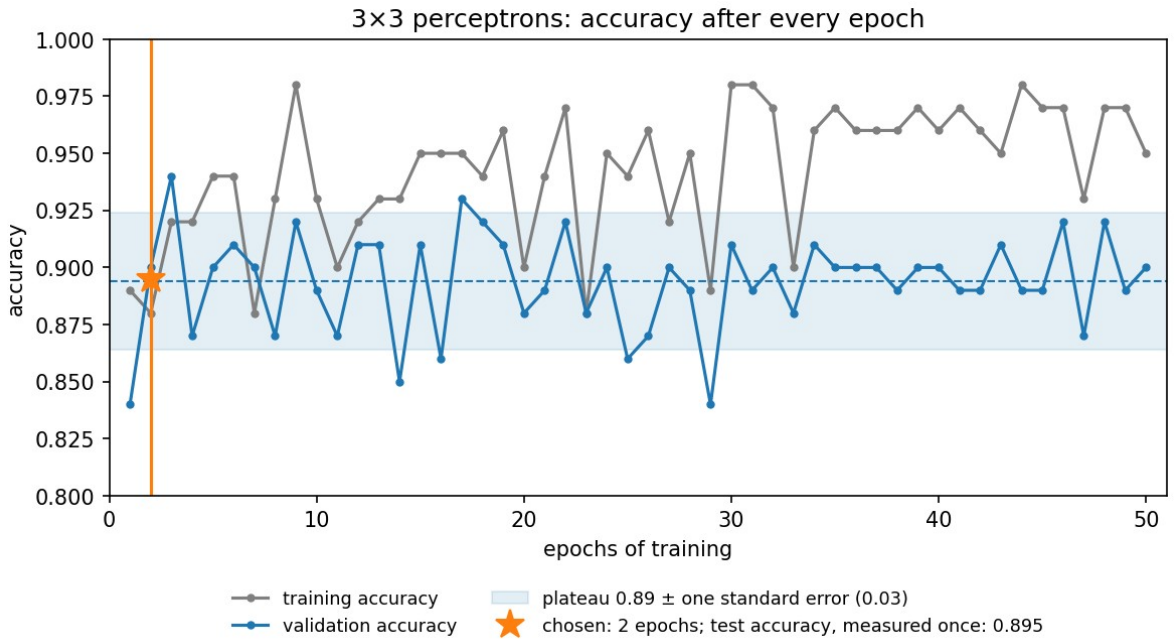


Training: fit the weights. Validation: make choices (epochs, size, features). Test: look once, at the very end; a test set used to guide choices becomes a second validation set.

See course.bioinfo.se/session08.html, 'Why three data sets?'.

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Choosing the number of epochs on the validation set

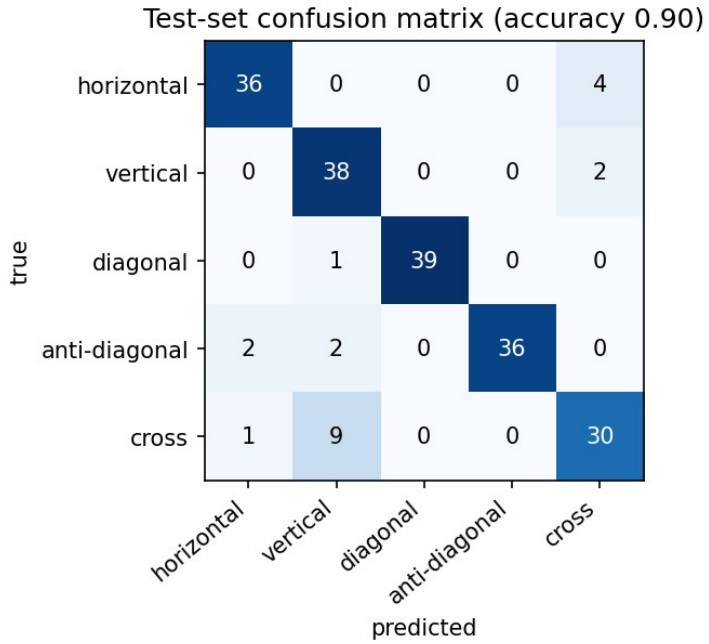


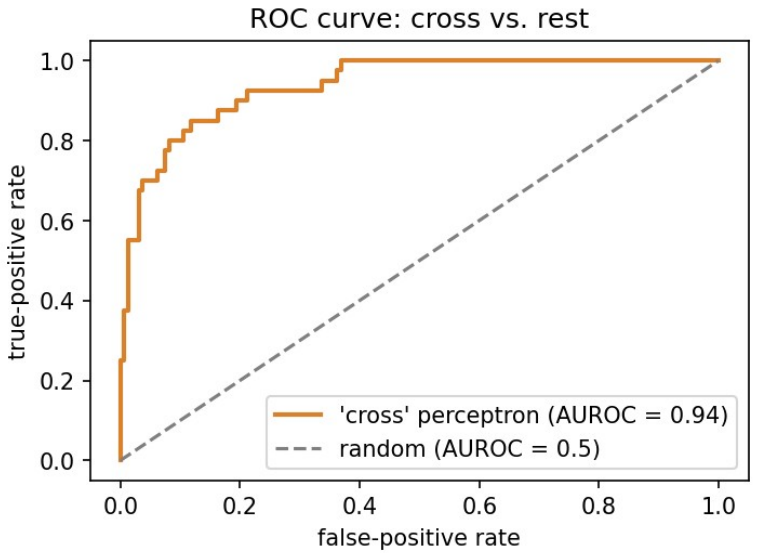
Validation accuracy is noisy (±0.03 with 100 examples) around a plateau of 0.89. The best single epoch (3, 0.94) is partly luck. Rule: the earliest epoch clearly on the plateau → 2 epochs. Test, once: 0.895.

Notebook, 'Training, validation and test sets, on the 3×3 problem'.

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The confusion matrix shows which mistakes



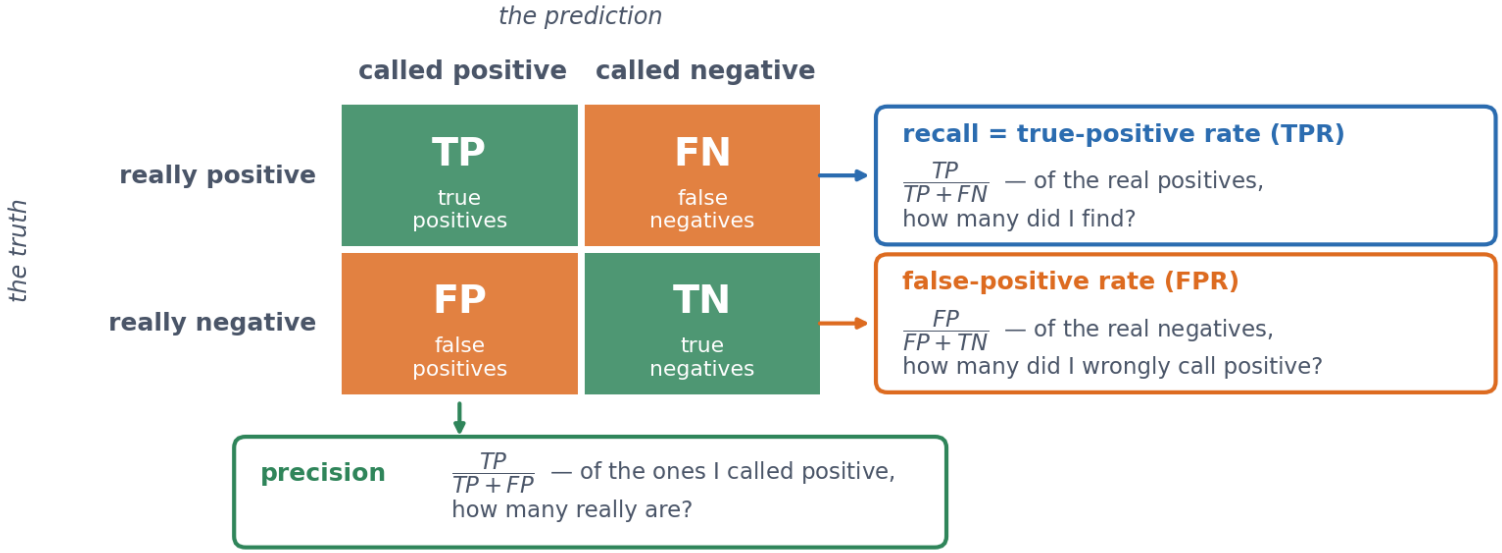


Left: almost all mistakes involve 'cross': 9 crosses called vertical, 4 horizontals and 2 verticals called cross. Right: sweeping the threshold on the cross perceptron's score traces the ROC curve.

Notebook: session08-ml-foundations-and-pytorch.ipynb.

Four counts, three measures

Precision is read down a column, recall across a row. The ROC curve plots TPR against FPR.



Rows are the truth, columns the prediction. Precision is read down a column; recall and the false-positive rate across a row.

See course.bioinfo.se/session08.html, 'Evaluating a classifier'.

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Precision and recall

- **precision = TP / (TP + FP)**
- **recall = TP / (TP + FN)**
- recall = TPR, the true-positive rate
- Screening: favour recall
- Expensive follow-up: favour precision

0.95

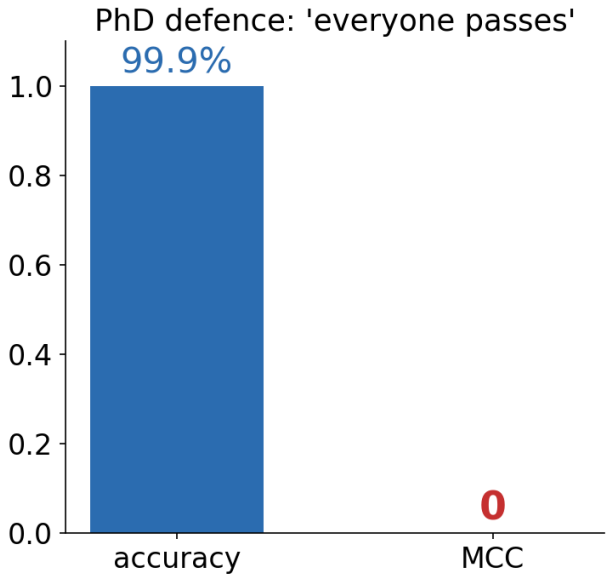
recall,
'vertical'

0.76

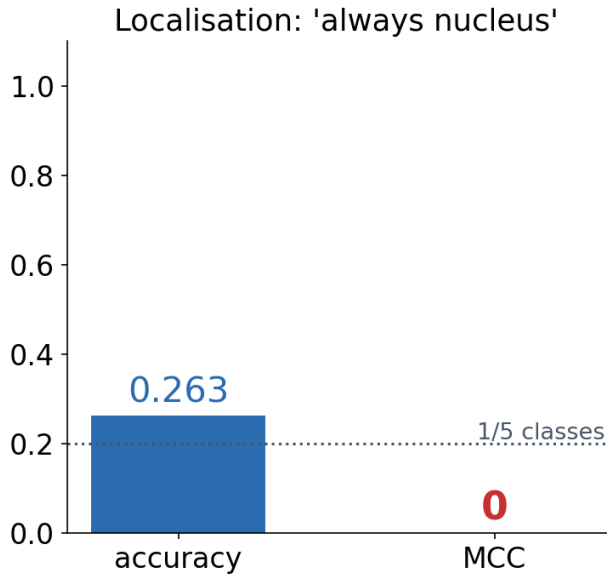
precision,
'vertical'

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High accuracy, no information



about 1 in 1000 fails



unbalanced classes: 424 / 497 / 140 / 488 / 338

Predicting that every PhD student passes is right 99.9% of the time. Always answering 'nucleus' on the unbalanced localisation data beats the 1-in-5 guess. Both have an MCC of exactly 0.

See course.bioinfo.se/session08.html, callout 'How useful is accuracy?' and 'Class imbalance first'.

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One number each: MCC and AUROC

- **MCC: all four counts, −1 to +1**
- 0 = chance; honest when classes are unbalanced
- **AUROC: P(a random positive outscores a random negative)**
- 0.5 = guessing, 1.0 = perfect ranking

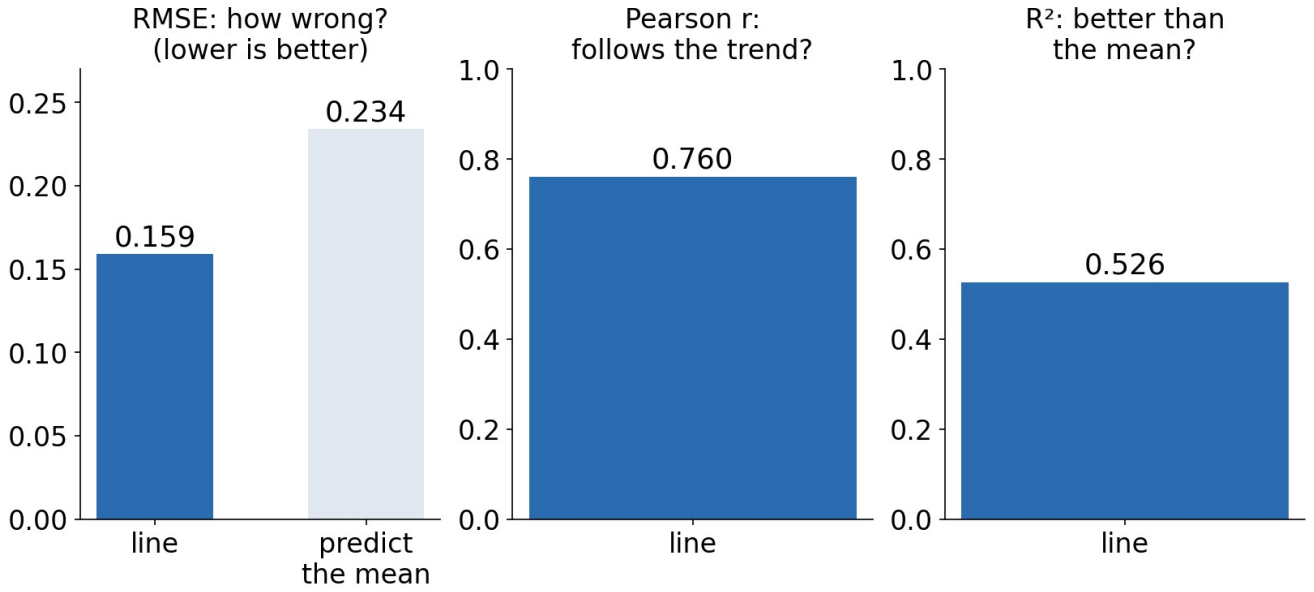
0.871
multi-class MCC,
3×3 test set

0.938
AUROC,
'cross'

See course.bioinfo.se/session08.html, 'Evaluating a classifier' and 'ROC curves'.

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Three numbers, three questions

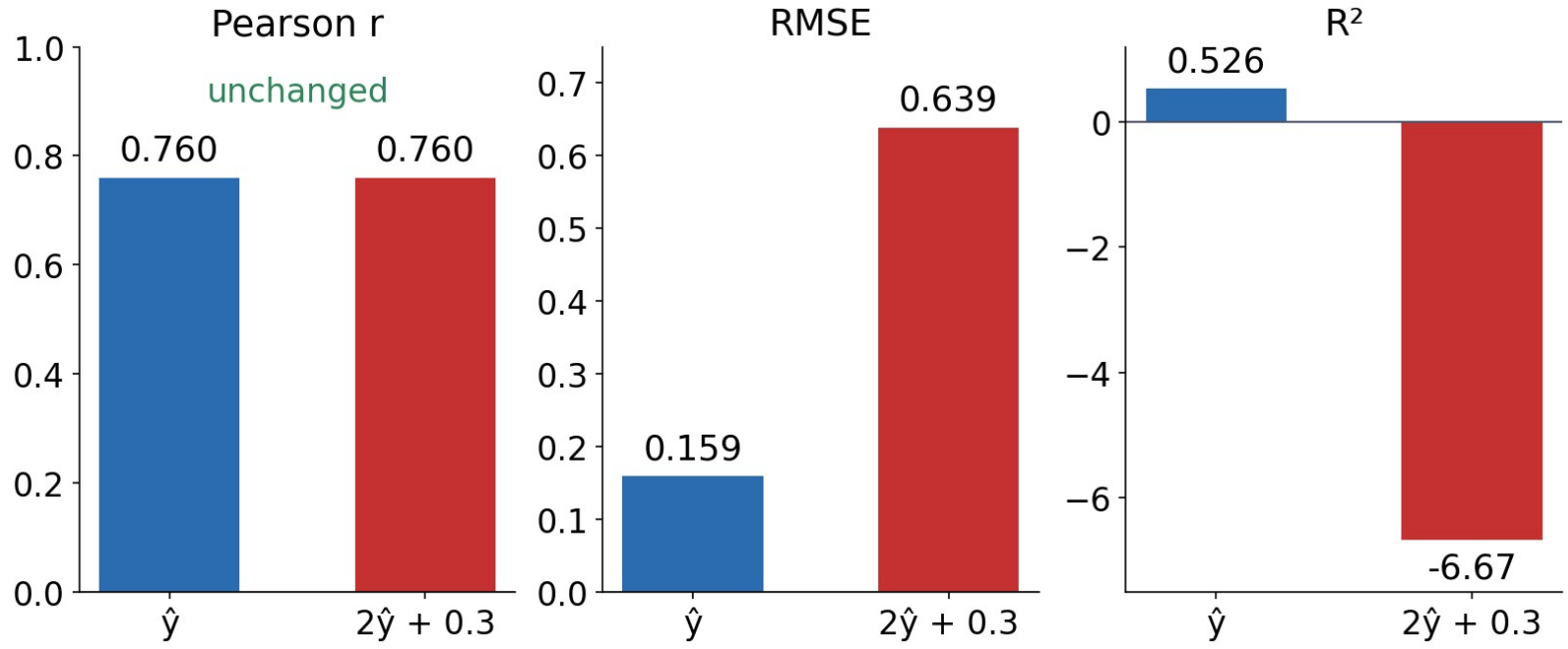


helix fraction from helix-former content: trained on 41 proteins, 14 held out

RMSE is in the target's units; predicting the training mean gives 0.234, the baseline to beat. Pearson r measures the trend; R^2 compares with always predicting the mean (about half the variance explained).

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Correlation is not accuracy

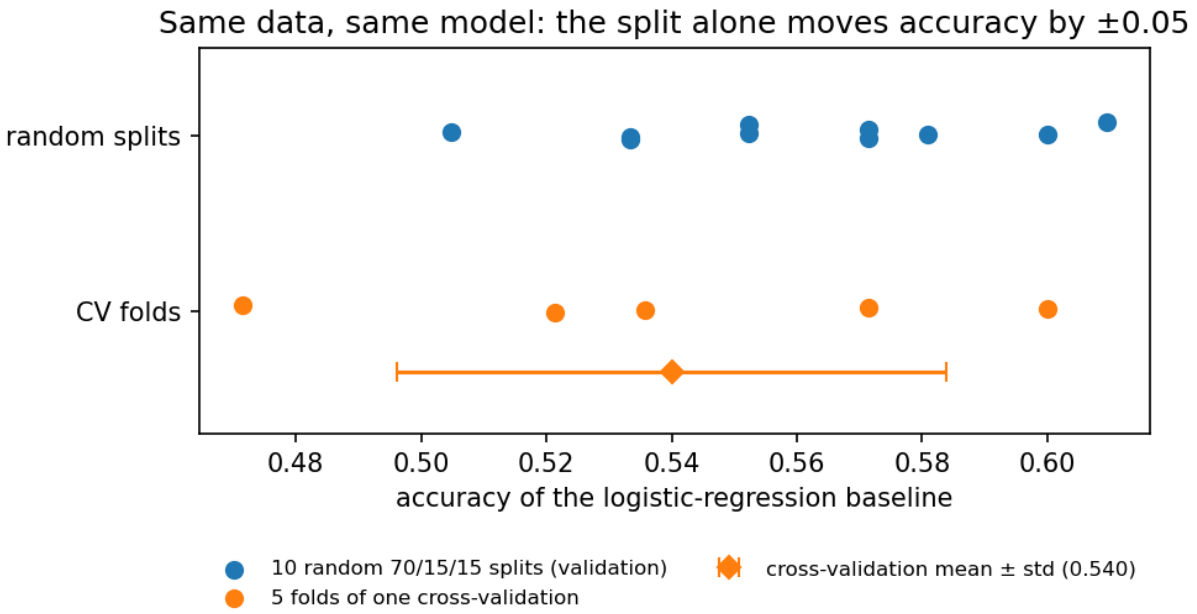


The same predictions stretched and shifted ($2\hat{y} + 0.3$): the correlation does not change at all, while RMSE quadruples and R^2 falls far below the mean predictor. A common mistake in bioinformatics papers.

Notebook, 'Evaluating a regression model'.

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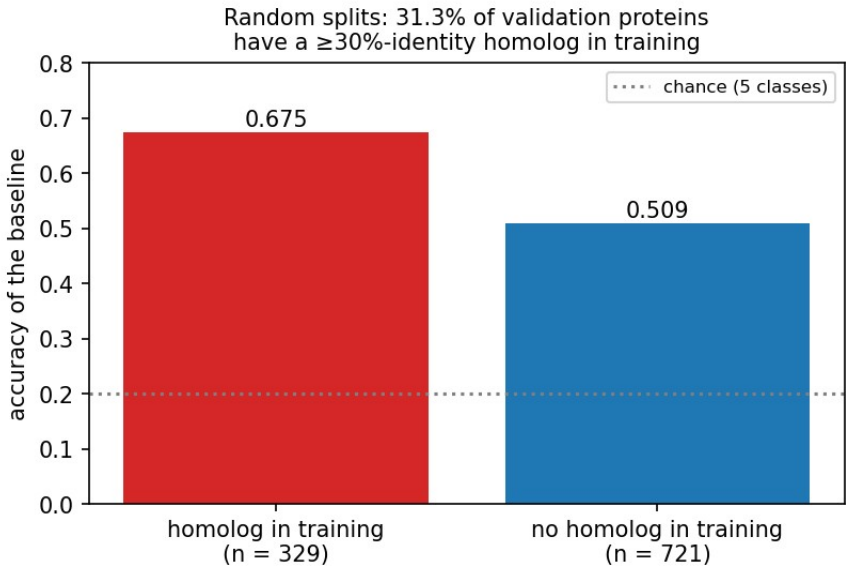
Same data, same model, different split

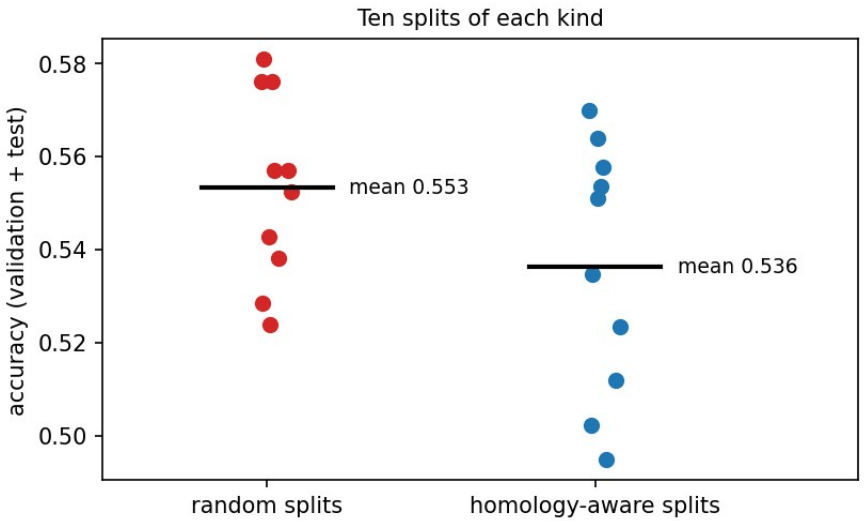


- **10 random splits: 0.505 to 0.610**
- 5-fold CV: 0.540 ± 0.044
- Every protein validates once
- At 5× the computing cost

Localisation: 700 human proteins, 5 classes, first 150 residues one-hot (3000 numbers).

Homologs on both sides of the split

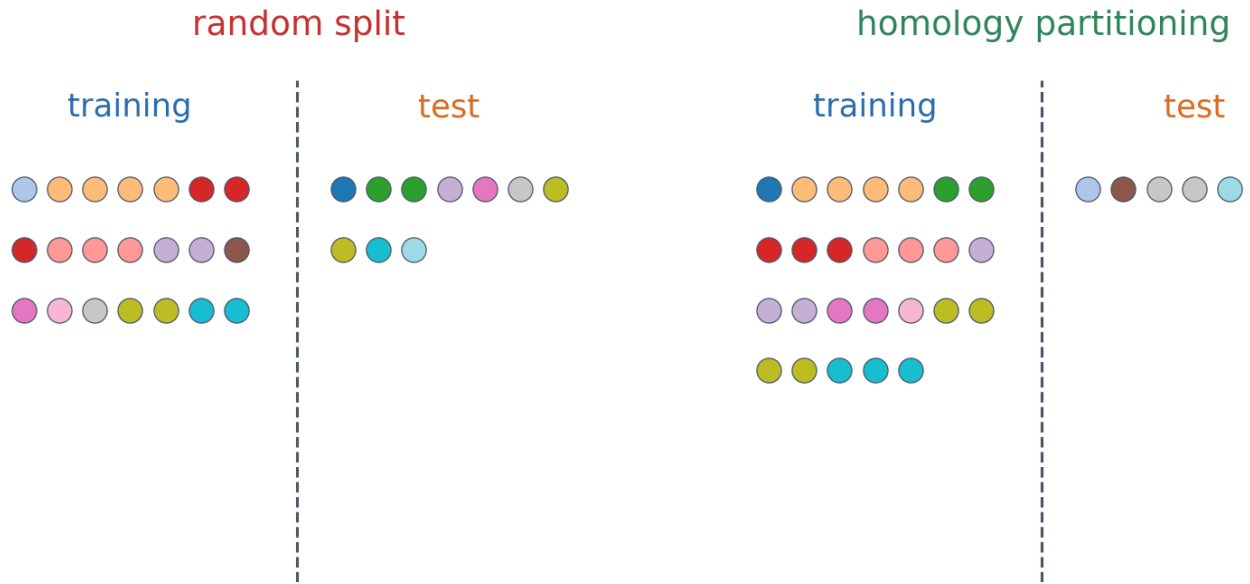




MMseqs2 at 30% identity: 700 proteins → 551 clusters. In random splits 31.3% of validation proteins have a homolog in training; those are right 67.5% of the time, the rest 50.9%.

Steinegger & Söding (2017), MMseqs2, Nat Biotechnol 35:1026.

Homology partitioning: split clusters, not proteins



one colour = one cluster of homologs (illustrative). Random: families on both sides. Partitioned: whole clusters move together.

Cluster first, then put whole clusters into training, validation or test. Over ten splits of each kind, baseline accuracy drops from 0.553 (random) to 0.536 (homology-aware). From today the course uses that split: 486 / 105 / 109 proteins.

Illustrative clusters. Numbers: 10 random and 10 homology-aware splits (notebook).

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Other ways test information leaks in

samples

homologs, or one patient's samples, on both sides

preprocessing

normalising or selecting features on all the data

evaluation

looking at the test set while choosing models

using the validation set to choose is not leakage: that is what it is for

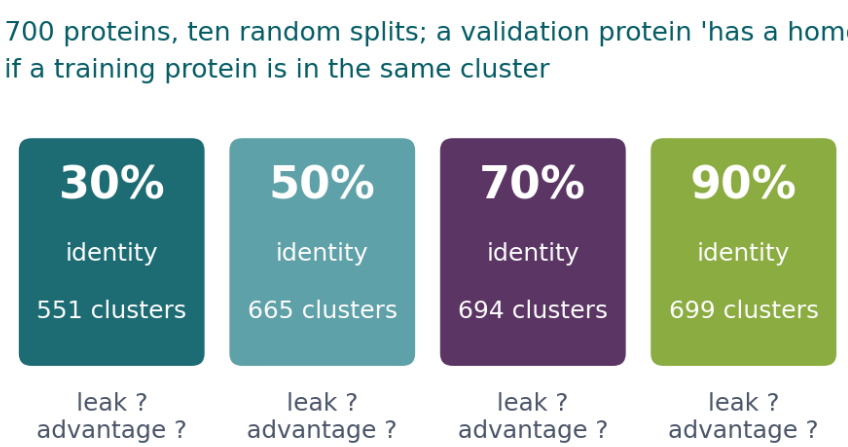
Also: batch effects (all tumours sequenced in one lab), time (tested on proteins annotated before the training data), and unrepresentative data. CASP's truly blind test (Session 6) is the model for honest evaluation.

See course.bioinfo.se/session08.html, 'Other ways biological data leaks'.

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What happens as the identity cutoff rises?

700 proteins clustered at 30%, 50%, 70% and 90% identity. As the cutoff rises: the share of validation proteins with a homolog in training, and their accuracy advantage?



At 30%: 31.3% leak; with a homolog 0.675 correct, without 0.509.

A Both fall; the advantage is gone at 50%

B The share falls, but the advantage grows

C Both stay about the same

D The share rises

Guess on your own

Scan the code, or go to

quiz.bioinfo.se/live

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What happens as the identity cutoff rises?

How did the room answer?

Halt the round when most have answered, then show the results.

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SciLifeLab

What happens as the identity cutoff rises?

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Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour

The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session08-discussion-2.ipynb

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What happens as the identity cutoff rises?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

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What happens as the identity cutoff rises?

Results, and who explains

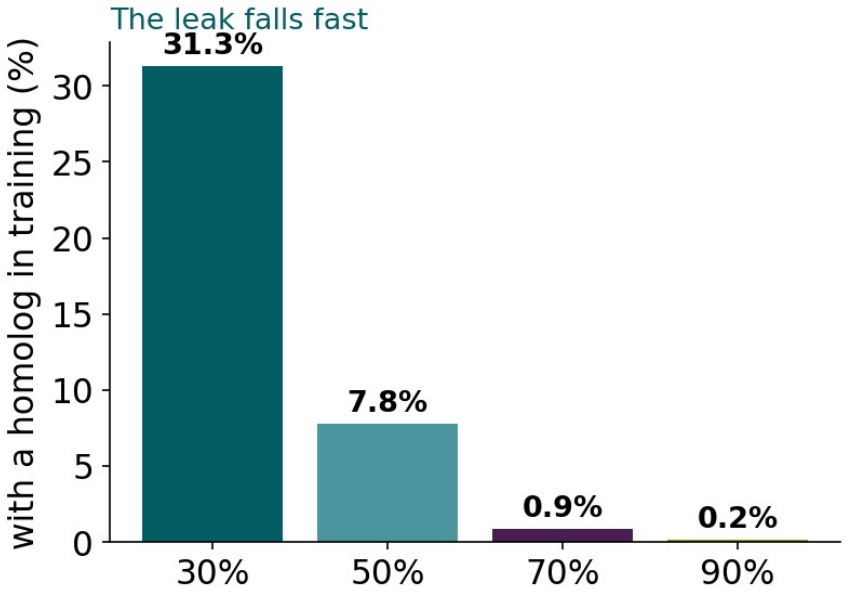
Both rounds side by side.

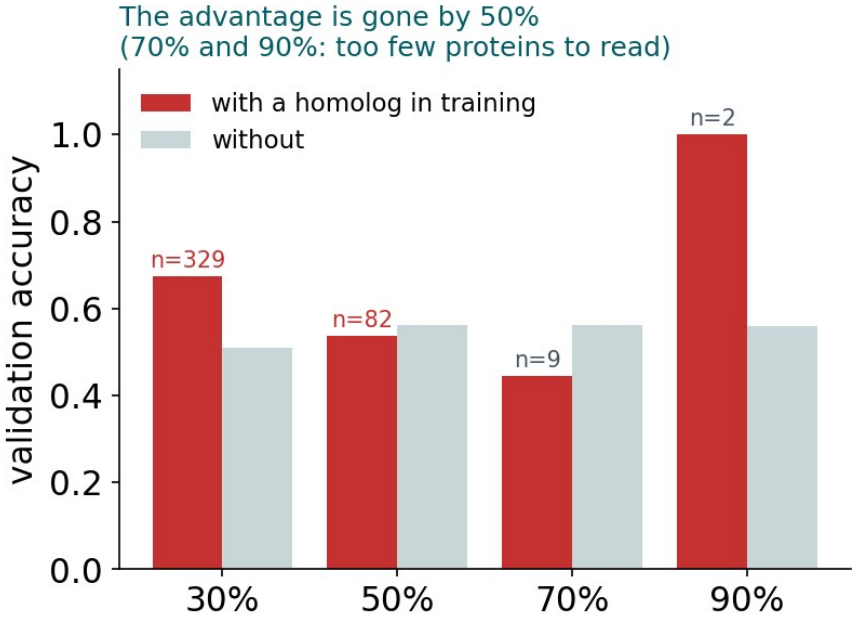
Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

To check it with code afterwards: discussion notebook 2 https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session08-discussion-2.ipynb

The answer: both fall; above 50% nothing is left to exploit





The share of validation proteins with a homolog in training falls from 31.3% at 30% identity to 7.8% at 50%, and their advantage disappears. At 70% and 90% there are too few such proteins to say anything. For this model the leak comes from distant relatives; that is why datasets are partitioned at 25-30%.

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BLOCK 2 DONE

Break

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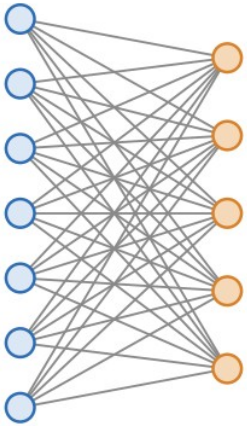
BLOCK 3 • ABOUT 30 MIN

Towards training networks

SciLifeLab

Connectivity decides the number of weights

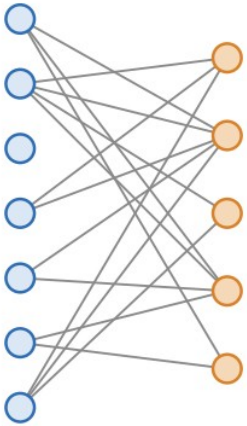
Fully connected



35 connections = 35 weights

Sessions 8-9: nn.Linear

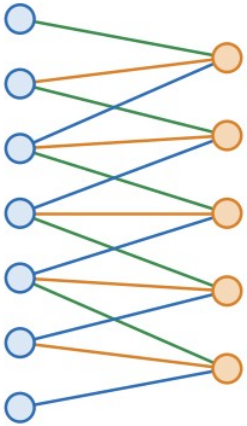
Sparsely connected



16 connections = 16 weights

closer to real brains

Local + shared weights



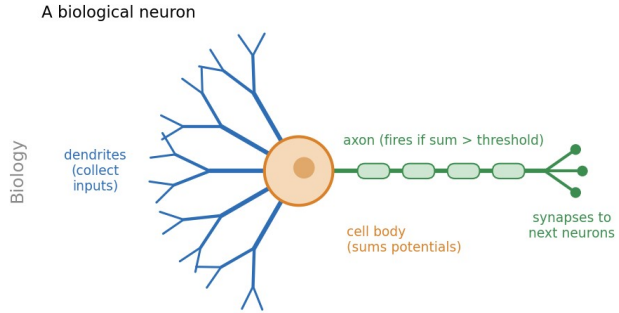
15 connections, 3 distinct weights

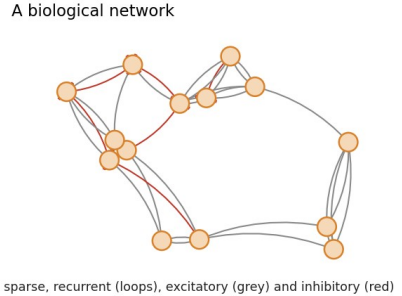
Session 10: convolution (same 3 colours reused)

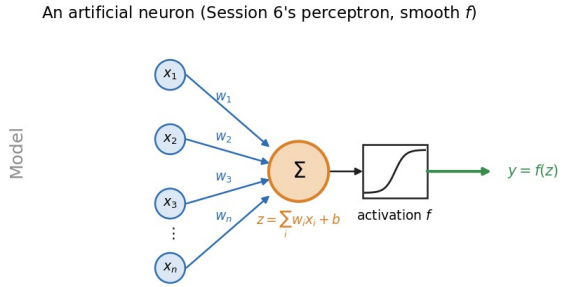
Fully connected (nn.Linear, Sessions 8-9): every input to every output, so the size grows with the input. Local with shared weights: convolution, Session 10's key idea.

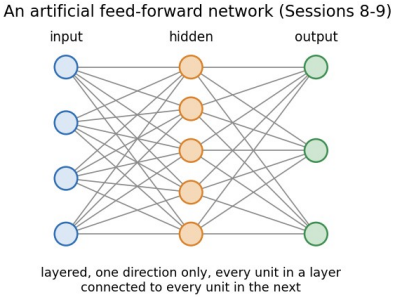
Figure drawn for this course.

Biological and artificial networks







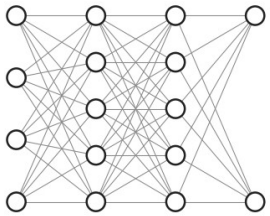


Real neurons spike, brains loop back and are sparsely connected. Artificial units output one number, and the networks of Sessions 8-10 are strictly feed-forward.

See course.bioinfo.se/session08.html, optional box 'Biological and artificial networks'.

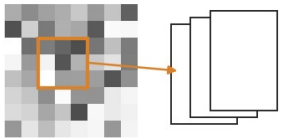
The architectures still to come

Feed-forward (MLP)
Sessions 8-9



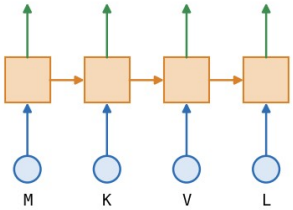
fixed-size input
e.g. localization, PHD

Convolutional (CNN)
Session 10



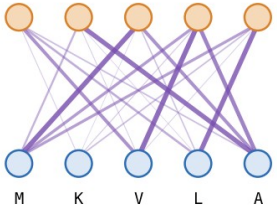
local patterns, shared weights
e.g. contact maps

Recurrent (RNN/LSTM)
Session 11



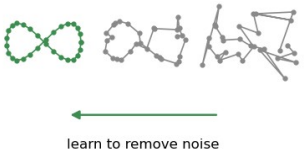
reads a sequence in order
e.g. signal peptides

Attention / Transformer
Session 11



every position looks at every other
e.g. ESM, AlphaFold

Generative (diffusion)
Session 12



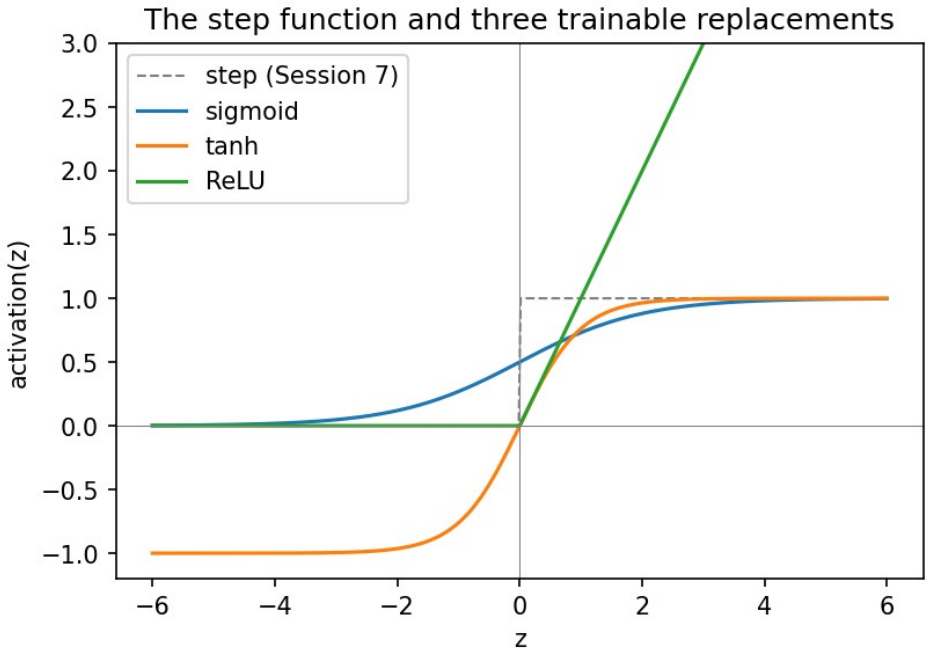
generates new examples
e.g. RFdiffusion design

Each builds a different assumption into its wiring: feed-forward (today, PHD), CNNs (contact maps, Session 10), RNNs and transformers (signal peptides, language models, Session 11), AlphaFold and diffusion models (Session 12).

See course.bioinfo.se/session08.html, optional box.

SciLifeLab

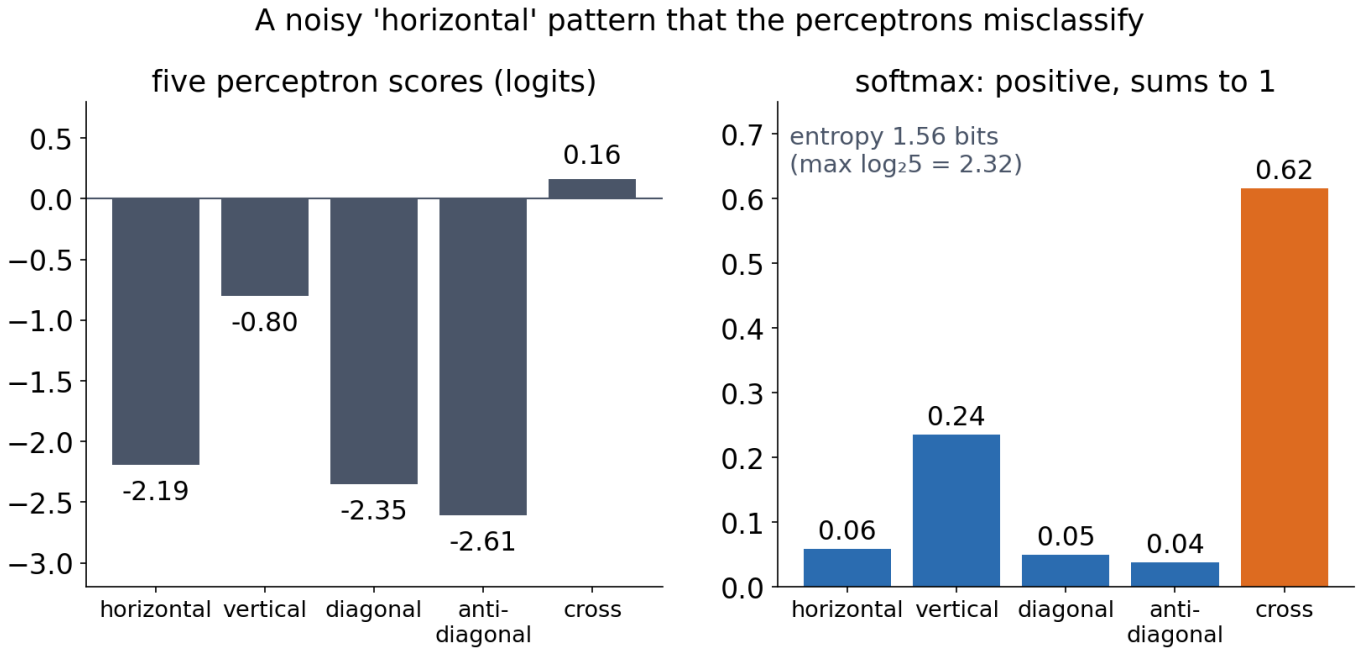
A step gives gradient descent nothing to follow



- **Step: derivative 0 almost everywhere**
- Sigmoid (0, 1): binary outputs
- tanh (−1, 1): centred on zero
- **ReLU: default for hidden layers**

See course.bioinfo.se/session08.html, 'From step functions to trainable activations'.

Softmax: K scores become a distribution



softmax$(z)_k = e^{z_k} / \sum_j e^{z_j}$. It never changes which class wins. These perceptrons were not trained with cross-entropy, so the numbers sum to 1 but are not reliable probabilities.

See course.bioinfo.se/session08.html, 'From binary to multi-class: softmax'.

SciLifeLab

In PyTorch the model outputs raw logits



apply softmax yourself only when you need normalised scores (e.g. a ROC curve)

nn.CrossEntropyLoss combines log-softmax with the negative log-likelihood in one numerically stable step, so the model should not apply softmax itself.

See course.bioinfo.se/session08.html, 'From binary to multi-class: softmax'; Session 9.

SciLifeLab

A model is a class: nn.Module

- **Layers as attributes in __init__**
- **forward(): input tensor → output tensor**
- The whole 3×3 classifier: nn.Linear(9, 5)
- Perceptron weights copied in → identical predictions

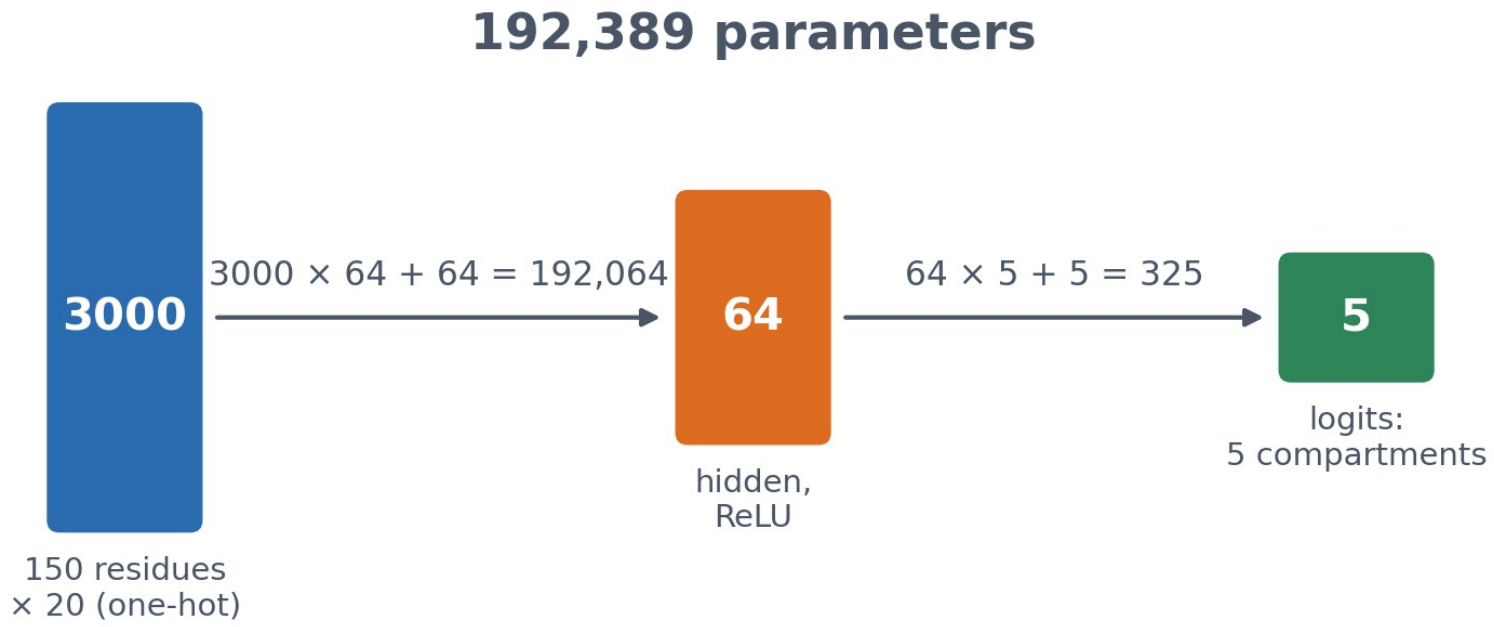
50

parameters:
5 × 9 + 5

Notebook: session08-ml-foundations-and-pytorch.ipynb. First real use of Session 2's class syntax.

SciLifeLab

The localisation network



3000 inputs (150 residues one-hot), a hidden layer of 64 ReLU units, 5 output logits. One forward pass on the homology-aware validation set confirms the shapes: 105 × 3000 in, 105 × 5 out.

Notebook: session08-ml-foundations-and-pytorch.ipynb.

SciLifeLab

Untrained, it scores what chance looks like

- **Random weights, homology-aware validation set**
- Not trained today
- **Session 9: backpropagation and the training loop, same metrics, same split**

0.181
accuracy

−0.027
MCC

0.474
AUROC

Notebook: session08-ml-foundations-and-pytorch.ipynb.

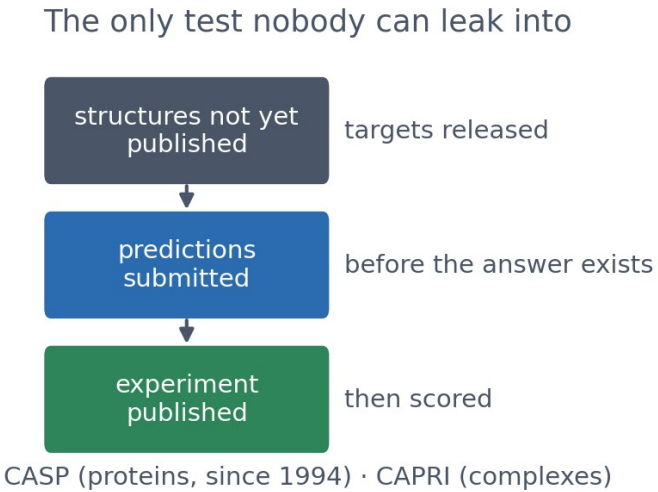
SciLifeLab

Leakage is not just a student mistake

17 scientific fields with leakage errors found

329 papers affected

8 types of leakage in their taxonomy

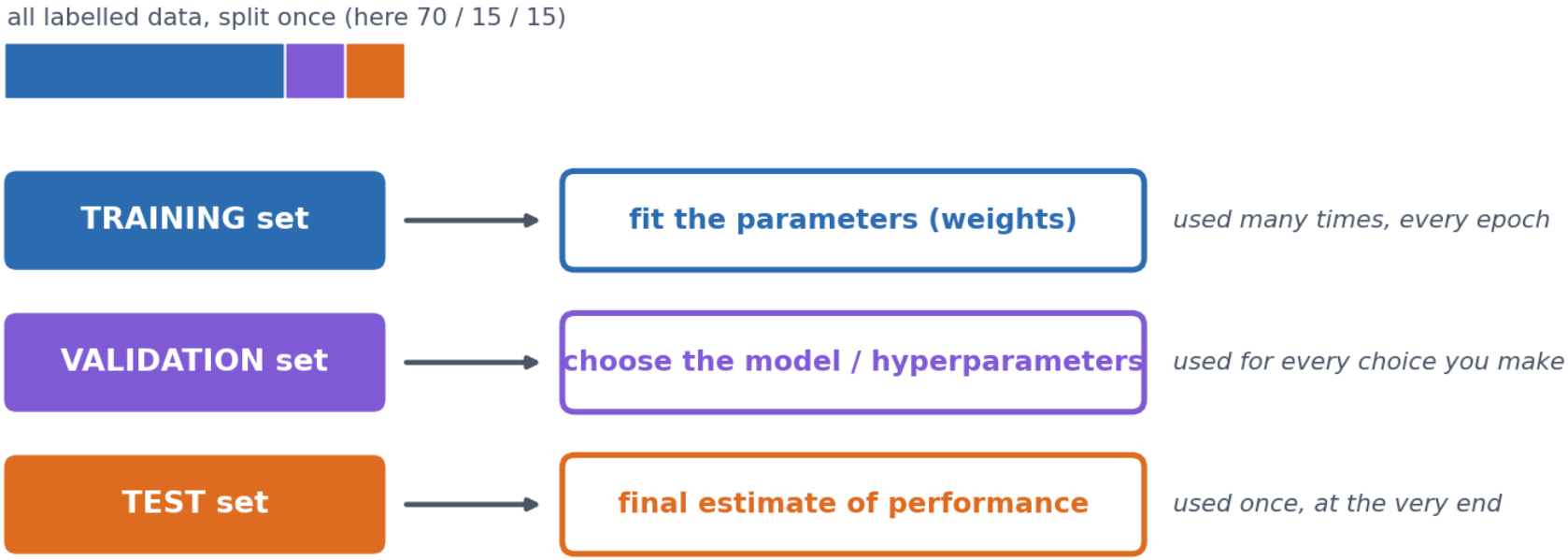


- Found across 17 fields of ML-based science
- Homologs in benchmarks, too
- Blind tests: the gold standard
- **How would you prove a model trained on millions of proteins never saw your test protein?**

Kapoor & Narayanan (2023), Patterns 4:100804; CASP (predictioncenter.org); CAPRI.

SciLifeLab

Evaluate on data the model has never seen



Pose the problem from the label. Split three ways and look at the test set once. Report MCC or precision and recall, not only accuracy; RMSE and R^2, not only correlation. Split proteins by homology, not at random.

See course.bioinfo.se/session08.html.

SciLifeLab

How do the parameter counts change with sequence length?

A one-hot sequence (L × 20) into nn.Linear(L·20, 64) or nn.Conv1d(20, 64, 7). From L = 150 to L = 1000?

one-hot sequence: L residues × 20 amino acids



`nn.Linear(L · 20, 64)`

every input to every one of 64 hidden units

`nn.Conv1d(20, 64, kernel_size=7)`

64 filters, each 7 residues wide, slid along the sequence

L = 150 → L = 1000: how do the parameter counts change?

A Both grow about seven times

B nn.Linear grows about 7×; nn.Conv1d stays the same

C Both stay the same

D nn.Conv1d grows; nn.Linear stays the same

Guess on your own

Scan the code, or go to
quiz.bioinfo.se/live

SciLifeLab

How do the parameter counts change with sequence length?

How did the room answer?

Halt the round when most have answered, then show the results.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

How do the parameter counts change with sequence length?

Fix it, run it, discuss

Open the notebook: scan the code

Fix the line marked FIX, then run the cells

Did it come out as you guessed? Discuss with your neighbour



The notebook: https://colab.research.google.com/github/ElofssonLab/kb8029-book/blob/main/notebooks/session08-discussion-3.ipynb

SciLifeLab

How do the parameter counts change with sequence length?

Answer again

Same question, after the discussion.

quiz.bioinfo.se/live

SciLifeLab

How do the parameter counts change with sequence length?

Results, and who explains

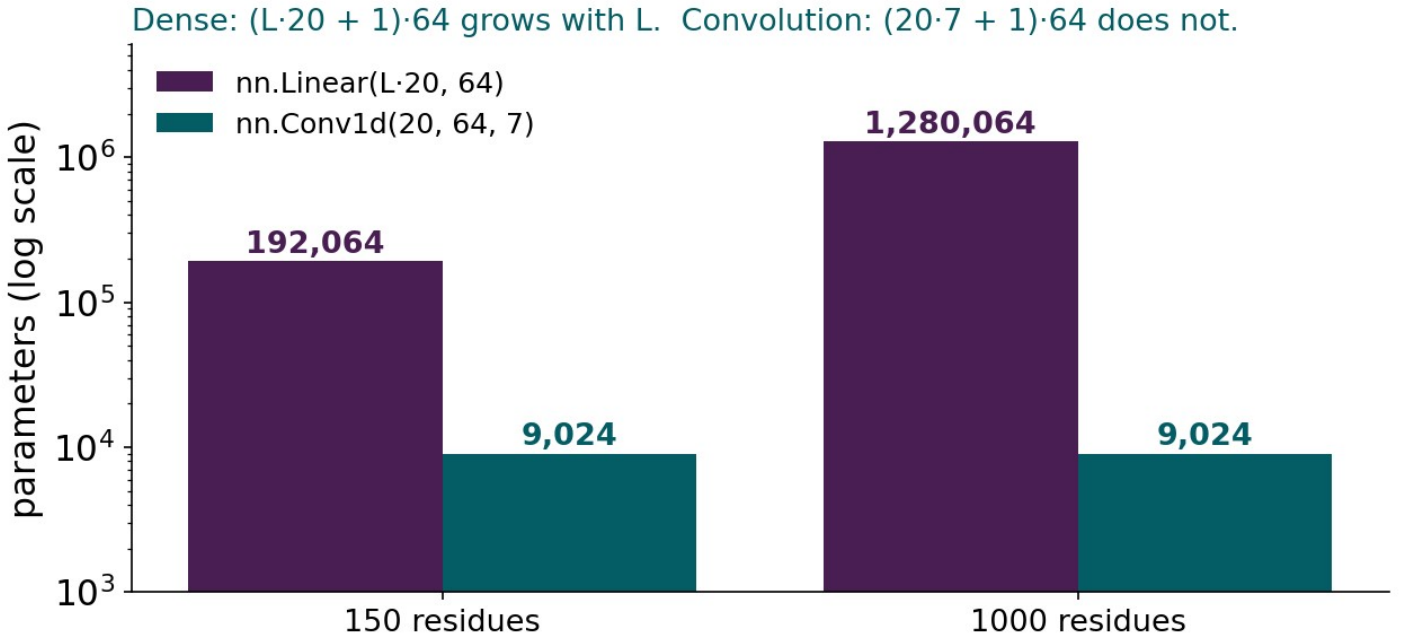
Both rounds side by side.

Draw two to explain: one of you says how you reasoned.

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

The answer: the dense layer grows with the sequence; the convolution does not



nn.Linear has (L·20 + 1)·64 parameters: 192,064 at 150 residues, 1,280,064 at 1000. nn.Conv1d has (20·7 + 1)·64 = 9,024 at any length, because one set of filters is slid along the whole sequence. That assumption, a motif means the same wherever it is, is Session 10's convolutional network.

SciLifeLab

In up to three words: what was the hardest part of today?

Posing the problem	Did it learn?	Towards networks
regression or classification	splits, metrics, homology leakage	softmax, nn.Module, parameters

Answer now

Scan the code, or go to
quiz.bioinfo.se/live



SciLifeLab

The hardest part of today

What the room found hardest

Show the results ▸ quiz.bioinfo.se/live/report

SciLifeLab

Session 9 next

This model, trained on the homology-aware split: backpropagation, cross-entropy, regularisation
And an honest comparison against the linear baseline